SporeERP — Master Role Architecture & Interaction Guide

Comprehensive System Guide: All 6 Roles, Operational Flowcharts, Wireframes & Security Matrix

PLATFORM CODE
SporeERP (Next.js 16 + MariaDB 11)

ROLES COVERED
6 Distinct Enterprise Roles

SECURITY STANDARD
NIST Least-Privilege RBAC

1. The 6 Enterprise System Roles Overview

Role Identifier	Display Name	Operational Scope	Key Responsibilities
FOUNDER_ADMIN	Co-Founder / Super Admin	Unrestricted Access	Real-time P&L, employee salaries, pricing overrides, audit logs, system settings.
FARM_MANAGER	Operations Manager	Facility & Production	Batch management, growing rooms, supplier purchases, customer orders, operational reports.
PRODUCTION_STAFF	Agronomy Staff	Growing Floor & Lab	Bed visualizer, bed colonization/fruiting status, contamination quarantine, harvest weighing.
INVENTORY_STAFF	Warehouse Keeper	Cold Storage & Warehouse	Stock-in/out, raw materials balance, FIFO freshness audits, spoilage write-offs.
OPERATIONS_STAFF	Fulfillment Agent	Packaging & Dispatch	Order packing, slip printing, delivery driver assignments, in-transit timestamps.
CUSTOMER	Retail / B2B Buyer	Storefront & Account	Browse varieties, order fresh mushrooms, wholesale tier pricing, live delivery tracking.

2. Master Role-to-Permissions Security Matrix (All 24 Atomic Permissions)

Module	Permission Code	Customer	Prod Staff	Inv Staff	Ops Staff	Farm Mgr	Founder
Users	users:view	❌	❌	❌	❌	👁 Read	✅ Full
Users	users:manage_roles	❌	❌	❌	❌	❌	✅ Full
Catalog	catalog:product:view	👁 Read	👁 Read	👁 Read	👁 Read	👁 Read	✅ Full
Catalog	catalog:product:manage	❌	❌	❌	❌	✅ Write	✅ Full

Catalog	catalog:pricing:override	❌	❌	❌	❌	❌	✅ Full
Orders	orders:create_own	✅ Yes	❌	❌	❌	❌	✅ Full
Orders	orders:view_all	❌	❌	👁️ Read	👁️ Read	👁️ Read	✅ Full
Orders	orders:update_status	❌	❌	❌	✅ Write	✅ Write	✅ Full
Orders	orders:cancel_refund	❌	❌	❌	❌	✅ Write	✅ Full
Inventory	inventory:view_stock	❌	👁️ Read	👁️ Read	👁️ Read	👁️ Read	✅ Full
Inventory	inventory:adjust_stock	❌	❌	✅ Write	❌	✅ Write	✅ Full
Inventory	inventory:write_off_spoil	❌	❌	✅ Write	❌	✅ Write	✅ Full
Production	production:batch:create	❌	✅ Write	❌	❌	✅ Write	✅ Full
Production	production:batch:update	❌	✅ Write	❌	❌	✅ Write	✅ Full
Production	production:beds:manage	❌	✅ Write	❌	❌	✅ Write	✅ Full
Harvest	harvest:log_entry	❌	✅ Write	👁️ Read	❌	✅ Write	✅ Full
Harvest	harvest:quality_grade	❌	✅ Write	👁️ Read	❌	✅ Write	✅ Full
Suppliers	suppliers:manage	❌	❌	👁️ Read	❌	✅ Write	✅ Full
Purchases	purchases:create	❌	❌	✅ Write	❌	✅ Write	✅ Full
Expenses	expenses:record	❌	❌	❌	❌	✅ Write	✅ Full
Finance	finance:pnl:view	❌	❌	❌	❌	❌	✅ Full
Audit	audit:logs:inspect	❌	❌	❌	❌	❌	✅ Full